

H.W/ 24-07-26.

① What are various problem which we have observed in our society. pick any five problems find out they can be solved them by data structures and algorithms.

Ans:- 1. Traffic congestion

- Heavy traffic causes delays and accidents.
- solution:- use graphs and shortest path algorithms (Dijkstra's algorithm) to find the fastest route and reduce traffic.

2. Hospital patient management:-

- problem: long waiting times and poor patient record management.

solution:- use queues to manage patient appointments and trees / databases to store and retrieve patient records quickly.

3. waste collections:-

problem:- garbage is not collected efficiently.

solution:- use graphs and routes optimization algorithms to find the shortest route for garbage collection vehicles.

4. online shopping delivery:-

problem :- Delayed product delivery

solution :- use graph and shortest path algorithm to select the best delivery route
use sorting algorithms to prioritize orders

5. emergency services (ambulance / fire)

problem :- Emergency vehicles take longer to reach the location.

solution :- use graphs, GPS andijkstra's algorithm to identify the quickest path and reduce response time.

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Graph [int first, int list